Day 2 handout

Day 2: Programming Fundamentals

PhD Course in Integrative Neurosciences - Introduction to Scientific Programming

Overview

Today we start writing Python. First we make sure everyone can **run** Python on their own computer, then we cover the foundations of Python programming that allow you to write powerful, flexible programs: 0. **Setting up and running Python** - install the tools, use the terminal, create an environment, run code four ways, read errors

1. **Variables, Data Structures, and Basic I/O** - How to store and organize data effectively
2. **Operators, Expressions, and Control Structures** - How to manipulate data and control program flow
3. **Functions and Modular Programming** - How to organize code into reusable, maintainable components

Today uses **pure Python** only (the standard library). Packages for numerical work are introduced from Day 4. If the session runs out of time, Part 3 continues at the start of Day 3.

By the end of this lecture, you'll have all the essential building blocks needed to write real scientific programs. Every program ever written consists of these components - it's how you combine them that distinguishes well-written, robust code from poorly-written, difficult-to-maintain code.

Part 0: Setting up and running Python

This part is a condensed reference. The full step-by-step guide for Windows, macOS and Linux, with troubleshooting, is `setup/setup_tutorial.md`, and the install checklist to do **before** class is `resources/before_day2.md`.

0.1 The three pieces

Piece	What it is	Example
Interpreter	the program that reads and runs Python code	`python`
Editor	where you write code, which is just text	VS Code

Piece	What it is	Example
Terminal	where you type commands to run programs and manage environments	Anaconda Prompt (Windows), Terminal (macOS, Linux)

Almost every setup problem is about these three not finding each other: the terminal cannot find Python, or the editor runs a different Python than the one you expect.

0.2 Installing

1. **Python with conda (Miniconda):** installs Python and `conda`, a tool for environments. On Windows always use **Anaconda Prompt** afterwards.
2. **VS Code** (https://code.visualstudio.com/) with the **Python** extension.
3. Check in a terminal: `python --version` and `conda --version` both print a version.

0.3 The terminal

The terminal always has a **current folder**. Files and scripts are found relative to it, so most "file not found" errors mean you are in the wrong folder.

What you want	Windows (Anaconda Prompt)	macOS / Linux
Where am I?	`cd`	`pwd`
List files	`dir`	`ls`
Go into a folder / up one	`cd name` / `cd ..`	`cd name` / `cd ..`
Make a folder	`mkdir name`	`mkdir name`
Which Python?	`where python`	`which python`
Stop a program	Ctrl+C	Ctrl+C

Press **Tab** to complete names and the **up arrow** to repeat the previous command.

0.4 Environments

An **environment** is an isolated workspace with its own Python and packages. The name of the active one appears in brackets at the start of the prompt.

```
conda env create -f environment-base.yml     # once
conda activate scientific-programming        # every time you start working
python --version                             # Python 3.11.x
conda deactivate                             # leave it
```

Without conda: `python -m venv .venv`, activate it (`.venv\Scripts\activate` on Windows, `source .venv/bin/activate` elsewhere), then `pip install -r requirements-base.txt`. The environment is minimal on purpose; Day 3 adds JupyterLab, and from Day 4 it is extended with the scientific packages.

0.5 Four ways to run Python

1. **Interactive prompt:** type `python` (or `ipython`) and then code lines; `exit()` to leave. Quick tests, nothing is saved.
2. **Script from the terminal:** `cd` to its folder, then `python hello.py`.
3. **VS Code:** open the folder, select the interpreter (Command Palette, **Python: Select Interpreter**, the `scientific-programming` one), click the play button.
4. **Notebook:** code, results and text in one document. We use Jupyter from Day 3.

	Interactive prompt	Script	Notebook
Saved as a file	no	yes	yes
Good for	quick tests	programs you rerun	exploring, documenting
Starts fresh each run	n/a	yes	no (the kernel keeps variables)

0.6 Reading an error message

Read from the **bottom**: the last line is the error type and what went wrong; the line above gives the file and line number; then look at that line.

```
Traceback (most recent call last):
  File "broken_2.py", line 4, in <module>
    print("Twice as many:", neuron_cont * 2)
NameError: name 'neuron_cont' is not defined
```

Message	Usually means	Try
`SyntaxError`	not valid Python (missing `:`, bracket or quote)	the line shown and the one before it
`NameError`	misspelled name, or used before it is defined	compare spelling, run earlier lines first
`ModuleNotFoundError`	wrong module name, or not installed in this environment	check spelling and that the environment is active
`FileNotFoundError`	Python cannot find the file from the current folder	`cd` to the right folder
`IndentationError`	inconsistent indentation	4 spaces per level
`TypeError`	the operation does not fit the types, e.g. `"5" + 3`	print the values and their `type()`

0.7 Checkpoint and Plan B

You are ready for Part 1 when: `python --version` works, the environment is active, `setup/check_python.py` ends with `RESULT: ... ready`, `hello.py` runs from the terminal and from VS Code, and you fixed the broken scripts (Exercise 0.3). If you are stuck, use **Plan B**: Google Colab (https://colab.research.google.com/) runs Python in your browser, and we fix your computer after class.

Part 1: Variables, Data Structures, and Basic I/O

1.1 Variables

Variables are named containers that store data. In Python, you don't need to declare variable types explicitly - Python infers the type from the value you assign:

```
# Basic variable assignment
age = 25                    # integer
temperature = 36.6          # float
name = "Maria"              # string
is_valid = True             # boolean
```

Variable Naming Rules:

- Must start with a letter or underscore
- Can contain letters, numbers, and underscores
- Case-sensitive (`age` and `Age` are different variables)
- Cannot use Python keywords (`if`, `for`, `while`, etc.)

> [!info] **Best Practices (PEP 8):**
>
> - Use lowercase with underscores for variable names: `neuron_count`, `reaction_time`
> - Use descriptive names: `spike_threshold` rather than `st`
> - Avoid single-letter names except for counters (`i`, `j`, `k`)

1.2 Basic Data Types

Python provides several built-in data types:

```
# Numeric types
integer_num = 42                    # int - whole numbers
float_num = 3.14159                 # float - decimal numbers
complex_num = 2 + 3j                # complex - rarely used in basic programming

# Text type
text = "Hello, neuroscience!"       # str - strings

# Boolean type
condition = True                    # bool - True or False (note capitalization)

# None type
empty_value = None                  # NoneType - represents absence of value
```

1.3 Type Conversion

You can convert between types using built-in conversion functions:

```
        # String to number
        age_str = "25"
        age_int = int(age_str)              # Converts to integer
        age_float = float(age_str)          # Converts to float

        # Number to string
        count = 100
        count_str = str(count)              # Converts to "100"

        # Important: Some conversions may fail
        invalid = "hello"
        # number = int(invalid)             # This would raise a ValueError!
```

1.4 Compound Data Types

Beyond simple types, Python provides structures to store collections of data:

Lists: *Ordered, Mutable Collections*

Lists are the most versatile data structure, allowing you to store ordered sequences of items:

```
        # Creating lists
        spike_times = [12.3, 15.7, 18.2, 22.1]      # List of floats
        brain_regions = ["hippocampus", "cortex", "thalamus"]  # List of
strings
        mixed_data = [1, "two", 3.0, True]          # Lists can contain mixed
types

        # Accessing elements (zero-indexed)
        first_spike = spike_times[0]                # 12.3
        last_region = brain_regions[-1]             # "thalamus" (negative
indexing from end)

        # Modifying lists
        spike_times.append(25.4)                    # Add element to end
        spike_times[0] = 10.0                       # Modify element
        spike_times.remove(15.7)                    # Remove specific value

        # List slicing
        first_three = spike_times[0:3]              # Elements 0, 1, 2
        last_two = spike_times[-2:]                 # Last two elements
```

Tuples: *Ordered, Immutable Collections*

Tuples are like lists but cannot be modified after creation:

```
        # Creating tuples
        coordinates = (10.5, 20.3, 30.1)            # Tuple of coordinates
        single_item = (42,)                         # Note: comma needed for
single item
```

```
        # Accessing elements (same as lists)
        x = coordinates[0]                          # 10.5
        y = coordinates[1]                          # 20.3

        # Tuples are immutable
        # coordinates[0] = 15.0                     # This would raise an
error!

        # Use tuples when data shouldn't change
        electrode_position = (100, 200, 300)        # Position should be fixed
```

Dictionaries: *Key-Value Pairs*

Dictionaries store data as key-value pairs, allowing fast lookups:

```
        # Creating dictionaries
        experiment_data = {
            "subject_id": "S001",
            "age": 28,
            "condition": "control",
            "score": 85.5
        }

        # Accessing values by key
        subject = experiment_data["subject_id"]     # "S001"
        age = experiment_data["age"]                # 28

        # Adding or modifying entries
        experiment_data["gender"] = "F"             # Add new key-value pair
        experiment_data["score"] = 90.0             # Update existing value

        # Checking if key exists
        if "condition" in experiment_data:
            print(f"Condition: {experiment_data['condition']}")

        # Getting all keys and values
        keys = experiment_data.keys()               # dict_keys(['subject_id',
'age', ...])
        values = experiment_data.values()           # dict_values(['S001', 28,
...])
```

Sets: *Unordered, Unique Collections*

Sets store unique values with no duplicates:

```
        # Creating sets
        neuron_types = {"pyramidal", "interneuron", "basket"}
        spike_counts = {1, 2, 3, 2, 1, 4}          # Duplicates automatically
removed → {1, 2, 3, 4}

        # Set operations
```

```
        all_neurons = {"pyramidal", "interneuron", "basket", "chandelier"}
        recorded_neurons = {"pyramidal", "interneuron"}

        common = all_neurons & recorded_neurons     # Intersection
        different = all_neurons - recorded_neurons  # Difference
        combined = all_neurons | recorded_neurons   # Union
```

1.5 Basic Input and Output

Reading User Input

The `input()` function reads text from the user:

```
        # Reading string input
        name = input("What is your name? ")
        print(f"Hello, {name}!")

        # Reading numeric input (requires conversion)
        age_str = input("What is your age? ")
        age = int(age_str)                          # Convert string to
integer

        # Modern approach: combine input and conversion
        age = int(input("What is your age? "))
```

Displaying Output

The `print()` function displays information:

```
        # Basic printing
        print("Hello, world!")

        # Printing multiple items (separated by space by default)
        name = "Ana"
        age = 25
        print("Name:", name, "Age:", age)           # Name: Ana Age: 25

        # Modern f-string formatting (Python 3.6+)
        print(f"Name: {name}, Age: {age}")          # Name: Ana, Age: 25

        # F-strings with expressions
        voltage = 70
        print(f"The membrane potential is {voltage}mV")
        print(f"Twice the voltage: {voltage * 2}mV")

        # Formatting numbers
        pi = 3.14159265359
        print(f"Pi to 2 decimals: {pi:.2f}")        # Pi to 2 decimals: 3.14
        print(f"Pi in scientific notation: {pi:.2e}")  # Pi in scientific
notation: 3.14e+00
```

Part 2: Operators, Expressions, and Control Structures

2.1 Understanding Operators and Expressions

An **operator** is a symbol that performs an operation on one or more **operands** (values or variables). Together, they form an **expression** that evaluates to a result.

Operators can be classified by:

1. **Number of operands**: Unary (one operand), Binary (two operands), Ternary (three operands)
2. **Type of operation**: Arithmetic, Logical, Comparison, etc.
3. **Type of result**: Numeric, Boolean, etc.

2.2 Arithmetic Operators

Arithmetic operators perform mathematical operations:

Operator	Name	Example	Result	Description
`**`	Power	`2 ** 3`	`8`	Exponentiation
`+`	Unary plus	`+5`	`5`	Positive sign
`-`	Unary minus	`-5`	`-5`	Negative sign
`*`	Multiplication	`4 * 5`	`20`	Multiplication
`/`	Division	`7 / 2`	`3.5`	True division (always returns float)
`//`	Floor division	`7 // 2`	`3`	Integer division (rounds down)
`%`	Modulo	`7 % 3`	`1`	Remainder of division
`+`	Addition	`3 + 4`	`7`	Addition
`-`	Subtraction	`10 - 3`	`7`	Subtraction

Operator Precedence (highest to lowest):

1. ** (power)
2. +, - (unary)
3. *, /, //, %
4. +, - (binary)

```
        # Examples demonstrating precedence
        result = 2 + 3 * 4              # 14 (multiplication before addition)
        result = (2 + 3) * 4            # 20 (parentheses override precedence)
        result = 2 ** 3 ** 2            # 512 (right-to-left: 2 ** (3 ** 2))
        result = -5 ** 2                # -25 (unary minus has lower
precedence than **)
        result = (-5) ** 2              # 25 (parentheses change meaning)
```

Compound Assignment Operators: These operators combine arithmetic operations with assignment:

```
        a = 10
        a += 5          # Equivalent to: a = a + 5  → a is now 15
        a -= 3          # Equivalent to: a = a - 3  → a is now 12
```

```
a *= 2          # Equivalent to: a = a * 2  → a is now 24
a /= 4          # Equivalent to: a = a / 4  → a is now 6.0
a //= 2         # Equivalent to: a = a // 2 → a is now 3.0
a %= 2          # Equivalent to: a = a % 2  → a is now 1.0
a **= 3         # Equivalent to: a = a ** 3 → a is now 1.0
```

2.3 Arithmetic Expressions

Operators combine with variables and values to form more complex expressions:

```
# Scientific example: calculating distance
x1, y1 = 10.0, 20.0
x2, y2 = 30.0, 40.0

# Calculate Euclidean distance
distance = ((x2 - x1) ** 2 + (y2 - y1) ** 2) ** 0.5

# Neuroscience example: calculating firing rate
spike_count = 150
recording_duration = 10.0  # seconds
firing_rate = spike_count / recording_duration  # 15.0 Hz

# Use parentheses for clarity even when not strictly needed
mean = (a + b + c) / 3      # Clear that we're averaging three values
```

> [!info] **Best Practices:**
>
> - Use spaces around operators for readability: `a + b * c` rather than `a+b*c`
> - Use parentheses to make complex expressions clear
> - Break very long expressions across multiple lines

2.4 String Operators

Strings have special operators for concatenation and repetition:

```
# Concatenation with +
greeting = "Hello" + " " + "world"          # "Hello world"
full_name = "Ana" + " " + "Silva"           # "Ana Silva"

# Repetition with *
separator = "-" * 40                        # "----------------------------------------"
emphasis = "Important! " * 3                # "Important! Important! Important! "

# Modern approach: use f-strings instead of concatenation
first_name = "Ana"
last_name = "Silva"
full_name = f"{first_name} {last_name}"     # Cleaner and more flexible
```

```
        # F-strings can include expressions
        neuron_count = 42
        message = f"Recorded {neuron_count} neurons ({neuron_count * 2}
electrodes used)"
```

2.5 Boolean (Logical) Variables

Boolean variables represent truth values and are essential for decision-making:

```
        # Boolean values (note capitalization - must be True/False, not
true/false)
        is_active = True
        is_inhibited = False

        # Comparison operations return booleans
        age = 25
        is_adult = age >= 18                        # True
        is_teenager = 13 <= age < 20                # False

        # Use booleans in conditions
        if is_active:
            print("Neuron is firing")
```

2.6 Comparison Operators

Comparison operators compare values and return boolean results:

Operator	Name	Example	Result	Description
`<`	Less than	`3 < 5`	`True`	Left value is smaller
`<=`	Less than or equal	`5 <= 5`	`True`	Left value is smaller or equal
`>`	Greater than	`3 > 5`	`False`	Left value is larger
`>=`	Greater than or equal	`5 >= 3`	`True`	Left value is larger or equal
`==`	Equal to	`5 == 5`	`True`	Values are equal
`!=`	Not equal to	`5 != 3`	`True`	Values are different

```
        # Numeric comparisons
        voltage = -70
        is_resting = voltage == -70                 # True
        is_depolarized = voltage > -70              # False
        is_hyperpolarized = voltage < -70           # False

        # String comparisons (alphabetical/lexicographic order)
        "apple" < "banana"                          # True
        "Apple" < "apple"                           # True (uppercase comes
before lowercase)

        # Chained comparisons (very useful!)
        age = 25
        is_young_adult = 18 <= age < 30             # True - much cleaner
```

```
than: age >= 18 and age < 30

        # Be careful with floating-point comparisons
        a = 0.1 + 0.1 + 0.1
        b = 0.3
        print(a == b)                                   # Might be False due to
floating-point precision!
        print(abs(a - b) < 1e-10)                       # Better approach for
floats
```

Common Pitfall:

```
        # Don't confuse assignment (=) with comparison (==)
        x = 5       # Assignment: set x to 5
        x == 5      # Comparison: check if x equals 5

        # This is a common error:
        if x = 5:   # WRONG! This is a syntax error
            print("x is 5")

        if x == 5:  # CORRECT - this compares x to 5
            print("x is 5")
```

2.7 Logical Operators

Logical operators combine boolean values:

Operator	Name	Example	Result	Description
`not`	Negation	`not True`	`False`	Inverts boolean value
`and`	Conjunction	`True and False`	`False`	True only if both operands are True
`or`	Disjunction	`True or False`	`True`	True if at least one operand is True

Truth Tables:

```
        # AND truth table
        True and True    # True
        True and False   # False
        False and True   # False
        False and False  # False

        # OR truth table
        True or True     # True
        True or False    # True
        False or True    # True
        False or False   # False

        # NOT truth table
        not True         # False
        not False        # True
```

Operation	Input 1	Input 2	Output
AND	True	True	True
AND	True	False	False
AND	False	True	False
AND	False	False	False
OR	True	True	True
OR	True	False	True
OR	False	True	True
OR	False	False	False
NOT	True		False
NOT	False		True

Practical Examples:

```
        # Neuroscience example: check if neuron should fire
        voltage = -50
        threshold = -55
        is_excitatory = True

        should_fire = voltage > threshold and is_excitatory  # True

        # Multiple conditions
        age = 25
        has_consent = True
        is_healthy = True

        can_participate = age >= 18 and has_consent and is_healthy  # True

        # Using 'or' for alternatives
        response = "y"
        is_affirmative = response == "yes" or response == "y" or response ==
"Y"  # True

        # Complex logical expressions
        temperature = 37.5
        is_fevered = temperature > 37.0
        is_medicated = False

        needs_attention = (is_fevered and not is_medicated) or temperature >
39.0
```

Operator Precedence (all operators, highest to lowest):

1. ** (power)
2. +, - (unary)
3. *, /, //, %
4. +, - (binary)
    1. <, <=, >, >=, == , != (comparisons)
5. not

6. and
7. or

```
        # Precedence examples
        result = 5 + 3 > 7 and 10 < 20          # True (arithmetic, then
comparison, then logical)

        # Use parentheses for clarity!
        result = (5 + 3 > 7) and (10 < 20)      # Same result but much clearer

        # Complex expression
        i = 0
        j = 10
        condition = i < j and j >= 5 or i > -1  # True - but hard to read!
        condition = ((i < j) and (j >= 5)) or (i > -1)  # Same thing, much
clearer
```

2.8 Control Structures

Control structures allow programs to make decisions and repeat actions. Without them, programs would be linear - executing every line in order from start to finish.

2.8.1 Branching: The `if` Statement

The `if` statement executes code conditionally based on boolean expressions.

Basic `if` Structure:

```
        # Simple if statement
        age = 25
        if age >= 18:
            print("You are an adult")
            print("You can participate in the study")
        # Code continues here regardless
```

`if-else` Structure:

```
        # Two alternatives
        age = int(input("What is your age? "))

        if age >= 100:
            print("You are already over 100!")
        else:
            years_to_go = 100 - age
            print(f"You will turn 100 in {years_to_go} years!")
```

Flowchart representation:

```
graph TD
    A[Start] --> B(Get age);
    B --> C{Age >= 100?};
    C -- Yes --> D[Over 100];
```

```
C -- No --> E[Calculate years];
D --> F[End];
E --> F;
```

`if-elif-else` Structure: For multiple conditions, use `elif` (else if):

```python
age = int(input("What is your age? "))

if age > 100:
    print("You are already over 100!")
elif age == 100:
    print("You are exactly 100, congratulations!")
else:
    years_to_go = 100 - age
    print(f"You will turn 100 in {years_to_go} years!")
```

> [!important] **Important Notes:**
>
> 1. **Indentation is crucial** - Python uses indentation (4 spaces recommended) to define code blocks
> 2. **Colons are required** after conditions
> 3. **Multiple statements** can be in each block
> 4. **elif vs multiple if** - they behave differently:

```python
# Using elif: only one block executes
score = 85
if score >= 90:
    print("Excellent")
elif score >= 80:
    print("Good")          # This prints
elif score >= 70:
    print("Satisfactory")  # This is skipped
# Output: "Good"

# Using multiple if: all matching conditions execute
score = 85
if score >= 90:
    print("Excellent")
if score >= 80:
    print("Good")          # This prints
if score >= 70:
    print("Satisfactory")  # This also prints!
# Output: "Good" and "Satisfactory"
```

Nested `if` Statements: You can nest `if` statements inside each other:

```python
voltage = -60
threshold = -55
refractory = False

if voltage > threshold:
    if not refractory:
        print("Neuron fires!")
```

```
            print("Action potential generated")
        else:
            print("In refractory period, cannot fire")
    else:
        print("Below threshold, no spike")
```

Neuroscience Example:

```
# Classifying neural response based on firing rate
firing_rate = 45.0  # spikes per second

if firing_rate > 100:
    response_type = "strong"
    print("High frequency response detected")
elif firing_rate > 50:
    response_type = "moderate"
    print("Moderate frequency response")
elif firing_rate > 10:
    response_type = "weak"
    print("Low frequency response")
else:
    response_type = "none"
    print("No significant response")

print(f"Classification: {response_type}")
```

2.8.2 Loops: `for` and `while`

Loops allow you to repeat actions multiple times without writing repetitive code.

The `for` Loop

Use `for` loops when you know how many times you want to repeat something, or when iterating through a sequence:

Basic `for` Loop with `range()`:

```
# Print numbers 0 to 4
for i in range(5):
    print(i)
# Output: 0, 1, 2, 3, 4

# range(start, stop, step)
for i in range(1, 11, 1):
    print(i)
# Output: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 (note: stop is exclusive!)

# Count by twos
for i in range(0, 10, 2):
    print(i)
# Output: 0, 2, 4, 6, 8
```

```python
# Count backwards
for i in range(10, 0, -1):
    print(i)
# Output: 10, 9, 8, 7, 6, 5, 4, 3, 2, 1
```

Important: `range(start, stop, step)` generates numbers from `start` up to **but not including** `stop`.

Iterating Through Sequences:

```python
# Iterate through a list
brain_regions = ["hippocampus", "cortex", "cerebellum", "thalamus"]
for region in brain_regions:
    print(f"Analyzing {region}")

# Iterate through a string
sequence = "ACGTACGT"
for nucleotide in sequence:
    print(nucleotide)

# Iterate with index using enumerate()
for index, region in enumerate(brain_regions):
    print(f"Region {index + 1}: {region}")
# Output:
# Region 1: hippocampus
# Region 2: cortex
# Region 3: cerebellum
# Region 4: thalamus
```

Practical Example - Sum of Squares:

```python
# Calculate sum of squares from 1 to n
n = int(input("Please enter a number: "))
sum_of_squares = 0

for i in range(1, n + 1):
    sum_of_squares += i ** 2

print(f"The sum of squares up to {n} is {sum_of_squares}")

# Example: if n = 5
# 1² + 2² + 3² + 4² + 5² = 1 + 4 + 9 + 16 + 25 = 55
```

Flowchart for sum of squares:

```
graph TD
    A[Start] --> B(Get n);
    B --> C[ssq = 0];
    C --> D[i = 1];
    D --> E{i <= n?};
    E -- Yes --> F[ssq += i²];
    F --> G[i += 1];
    G --> E;
```

```
    E -- No --> H[Print ssq];
    H --> I[End];
```

Nested `for` Loops:

```python
# Multiplication table
for n in range(1, 4):
    for m in range(1, n + 1):
        result = n ** m
        print(f"{n} to the power of {m} is {result}")
    print("=" * 40)

# Output:
# 1 to the power of 1 is 1
# ========================================
# 2 to the power of 1 is 2
# 2 to the power of 2 is 4
# ========================================
# 3 to the power of 1 is 3
# 3 to the power of 2 is 9
# 3 to the power of 3 is 27
# ========================================
```

The `while` Loop

Use `while` loops when you want to repeat something as long as a condition is true:

```python
# Basic while loop structure
while condition:
    # statements to repeat
    # (make sure condition eventually becomes False!)
```

Example - Find Minimum n:

```python
# Find minimum n where sum of squares exceeds x
x = int(input("Please enter a number: "))
sum_of_squares = 0
i = 1

while sum_of_squares <= x:
    sum_of_squares += i ** 2
    i += 1

print(f"The minimum n is: {i - 1}")
print(f"Sum of squares: {sum_of_squares}")

# Example: if x = 50
# 1² = 1, 1² + 2² = 5, 1² + 2² + 3² = 14, 1² + 2² + 3² + 4² = 30, 1² + 2² + 3² + 4² + 5² = 55
# Minimum n is 5 (sum = 55, which exceeds 50)
```

Flowchart for while loop:

```
graph TD
    A[Start] --> B(Get x);
    B --> C[ssq = 0];
    C --> D[i = 1];
    D --> E{ssq <= x?};
    E -- Yes --> F[ssq += i²];
    F --> G[i += 1];
    G --> E;
    E -- No --> H[Print i];
    H --> I[End];
```

Practical Example - User Validation:

```python
# Keep asking until valid input
password = ""
while password != "secret":
    password = input("Enter password: ")
    if password != "secret":
        print("Incorrect password, try again")

print("Access granted!")
```

> [!attention] **Danger: Infinite Loops** Make sure your condition eventually becomes `False`, or you'll create an infinite loop:
>
> ```python
> # DANGER - infinite loop!
> counter = 0
> while counter < 10:
>     print(counter)
>     # Oops! Forgot to increment counter
>     # This loop will never end!
>
> # CORRECT version:
> counter = 0
> while counter < 10:
>     print(counter)
>     counter += 1  # Don't forget this!
> ```

Loop Control: `break` and `continue`

Sometimes you need to exit a loop early or skip certain iterations:

The `break` Statement: Exits the loop immediately, skipping any remaining iterations:

```python
# Search for an element
search_letter = "r"
my_string = "This parrot is dead"

for letter in my_string:
    if letter == search_letter:
        print(f"Found '{search_letter}'!")
        break  # Exit loop once found
```

```
        else:
            # This executes if loop completes without break
            print(f"'{search_letter}' not found")

        # Multiple choice with limited attempts
        answer = 0
        max_attempts = 3

        for attempt in range(max_attempts):
            answer = int(input("Please enter your choice (1, 2, or 3): "))

            if 1 <= answer <= 3:
                print(f"Your answer was {answer}")
                break  # Valid answer, exit loop
            else:
                print(f"Invalid choice. {max_attempts - attempt - 1} attempts
remaining.")
        else:
            print("Too many invalid attempts. Goodbye!")
```

Flowchart with break:

```
graph TD
    A[Start loop] --> B{condition?};
    B -- No --> C[End loop];
    B -- Yes --> D{break condition?};
    D -- Yes --> E[EXIT LOOP];
    D -- No --> F[Continue];
    F --> B;
```

The `continue` Statement: Skips the rest of the current iteration and continues with the next one:

```
        # Process only integers from a mixed list
        my_list = [1, 6, 8, 12.4, "red", 9]
        my_int_list = []
        int_sum = 0

        for item in my_list:
            if not isinstance(item, int):
                continue  # Skip non-integers

            # These lines only execute for integers
            my_int_list.append(item)
            int_sum += item

        print(f"Integer list: {my_int_list}")
        print(f"Sum: {int_sum}")
        # Output:
        # Integer list: [1, 6, 8, 9]
        # Sum: 24

        # Skip negative numbers
```

```
for num in range(-5, 6):
    if num < 0:
        continue  # Skip negative numbers
    print(num)
# Output: 0, 1, 2, 3, 4, 5
```

Flowchart with continue:

```
graph TD
    A[Start loop] --> B{condition?};
    B -- No --> C[End loop];
    B -- Yes --> D{continue condition?};
    D -- Yes --> E[Skip to next];
    D -- No --> F[Execute rest];
    E --> B;
    F --> B;
```

> [!note] **Important Notes:**
>
> - `break` and `continue` only affect the innermost loop they're in
> - They should always be used with an `if` condition
> - Overuse can make code hard to follow - use judiciously

2.9 Exception Handling

Programs often encounter errors (called "exceptions"). Instead of crashing, we can handle these errors gracefully (we will cover this topic in more detail later):

Basic `try-except` Structure

```
# Without exception handling - program crashes
age_str = "twenty-five"
age = int(age_str)  # ValueError: invalid literal for int()
# Program terminates here!

# With exception handling - program continues
age_str = "twenty-five"
try:
    age = int(age_str)
    print(f"Your age is {age}")
except ValueError:
    print("You did not enter a valid number")
    age = None
# Program continues normally
```

Improved 100th Birthday Program:

```
my_age_str = input("What is your age? ")

try:
    my_age = int(my_age_str)
```

```
        if my_age > 100:
            print("You are already over 100!")
        elif my_age == 100:
            print("You are exactly 100, congratulations!")
        else:
            years_to_go = 100 - my_age
            print(f"You will turn 100 in {years_to_go} years!")

    except ValueError:
        print("You did not enter a valid number")
```

Giving Multiple Chances with `while` and `try-except`

```
    # Keep asking until valid input
    while True:
        my_age_str = input("What is your age? ")

        try:
            my_age = int(my_age_str)
            break  # Exit loop on successful conversion
        except ValueError:
            print("Invalid input. Please enter a number.")

    # Now process the valid age
    if my_age > 100:
        print("You are already over 100!")
    elif my_age == 100:
        print("You are exactly 100, congratulations!")
    else:
        years_to_go = 100 - my_age
        print(f"You will turn 100 in {years_to_go} years!")
```

Handling Multiple Exception Types

Different errors can be handled separately:

```
    try:
        numerator = int(input("Please enter the numerator: "))
        denominator = int(input("Please enter the denominator: "))

        quotient = numerator / denominator
        print(f"The result of {numerator} / {denominator} is: {quotient}")

    except ValueError:
        print("Incorrect input - please enter integers only!")

    except ZeroDivisionError:
        print("You cannot divide by zero!")

    except Exception as e:
```

```
    # Catch any other unexpected errors
    print(f"An unexpected error occurred: {e}")
```

Common Exception Types:

Exception	Description	Example
`ValueError`	Invalid value for operation	`int("hello")`
`ZeroDivisionError`	Division by zero	`5 / 0`
`TypeError`	Wrong type for operation	`"5" + 5`
`NameError`	Variable not defined	Using `count` before defining it
`IndexError`	Invalid index for sequence	`my_list[100]` on short list
`KeyError`	Invalid key for dictionary	`my_dict["nonexistent"]`
`FileNotFoundError`	File doesn't exist	Opening non-existent file
`IOError`	Input/output operation failed	Reading from closed file

The `try-except-else-finally` Structure:

```
try:
    # Code that might raise an exception
    file_content = open("data.txt").read()
    value = int(file_content)

except FileNotFoundError:
    # Executes if file not found
    print("Data file not found")
    value = 0

except ValueError:
    # Executes if content isn't a valid integer
    print("File contains invalid data")
    value = 0

else:
    # Executes only if NO exception occurred
    print(f"Successfully read value: {value}")

finally:
    # ALWAYS executes, whether exception occurred or not
    # Useful for cleanup (closing files, connections, etc.)
    print("Finished processing")
```

2.10 Importing Modules

Python's standard library and third-party packages provide powerful functionality. Use `import` to access them:

Import Methods

1. Import entire module:

```python
import math

# Use functions with module prefix
root = math.sqrt(16.0)
circle_area = math.pi * 2.0 ** 2
```

2. Import with alias (recommended for long names):

```python
import math as m

# Use shorter alias
root = m.sqrt(16.0)
circle_area = m.pi * 2.0 ** 2

# Other common aliases from the standard library:
import statistics as stats
import collections as col
```

3. Import specific functions:

```python
from math import sqrt, pi

# Use functions directly (no prefix needed)
root = sqrt(16.0)
circle_area = pi * 2.0 ** 2
```

4. Import everything (NOT recommended):

```python
from math import *

# Can use all math functions without prefix
# BUT: risk of name conflicts, hard to track where functions come from
root = sqrt(16.0)

# Danger: math has hundreds of names (pow, e, floor, ...). If you also
# define a variable or function with one of those names, one silently
# overwrites the other, and nobody reading the code can tell where
# sqrt() or pi came from.
```

Practical Example:

```python
import statistics

# Spike times from 10.0 to 19.9 ms in steps of 0.1 ms
# (range() only works with integers, so we scale and divide)
spike_times = [10.0 + i * 0.1 for i in range(100)]

# Calculate statistics
mean_time = statistics.mean(spike_times)
std_time = statistics.pstdev(spike_times)   # population standard
deviation

print(f"Spike times from {spike_times[0]:.1f} to {spike_times[-1]:.1f}
```

```
ms")
        print(f"Mean: {mean_time:.2f} ms, SD: {std_time:.2f} ms")
```

Useful Standard Library Modules: Many of your programs will start with some of these imports

```
# Mathematical functions and constants
import math

# Random numbers (seed it for reproducible results!)
import random
random.seed(42)
print(random.random())          # float in [0, 1)
print(random.randint(1, 6))     # integer between 1 and 6, inclusive

# Basic descriptive statistics
import statistics
print(statistics.median([3, 1, 4, 1, 5]))

# Specialised containers (counting things, default values)
from collections import Counter
print(Counter("abracadabra"))

# Dates and durations
from datetime import timedelta
print(timedelta(hours=1, minutes=30).total_seconds())

# Paths that work on any operating system
from pathlib import Path
data_file = Path("data") / "recording.txt"
print(data_file.suffix)         # .txt
```

Organising your imports: put all imports at the top of the file, in three groups separated by a blank line:

1. Standard library modules (`math`, `random`, `statistics`, ...), which come with Python
2. Third-party packages, which you install with `pip` (we cover this soon)
3. Your own local modules (other `.py` files in your project)

Besides the standard library, the Python ecosystem offers many third-party packages for scientific work. For now, the standard library is all you need.

Part 3: Functions and Modular Programming

3.1 What is a Function?

A **function** is a named block of reusable code that performs a specific task. Functions are the key to writing maintainable, understandable programs.

Anatomy of a function:

```
def function_name(parameters):
    """
    Docstring: describes what the function does.
    This is optional but highly recommended.
    """
    # Function body - the code that executes
    statement1
    statement2

    return result  # Optional - returns a value
```

Simple Example:

```
def greet(name):
    """Return a personalized greeting."""
    greeting = f"Hello, {name}!"
    print(greeting)
    print("It's nice to meet you")
    return greeting

# Call the function
message = greet("Ana")
# Output:
# Hello, Ana!
# It's nice to meet you
```

3.2 Why Use Functions?

Functions provide several critical benefits:

1. Avoid Repetition (DRY Principle: Don't Repeat Yourself)

```
# WITHOUT functions - repetitive and error-prone
print("Hello Abel")
print("It's nice to meet you")
print()
print("Hello Baker")
print("It's nice to meet you")
print()
print("Hello Charlie")
print("It's nice to meet you")

# WITH functions - clean and maintainable
def greet(name):
    print(f"Hello {name}")
    print("It's nice to meet you")
    print()
```

```
for person in ["Abel", "Baker", "Charlie"]:
    greet(person)
```

2. Make Code More Readable

```
# Hard to understand
data = [45, 52, 48, 51, 49, 55, 47]
result = sum(data) / len(data)
variance = sum((x - result) ** 2 for x in data) / len(data)
final = variance ** 0.5

# Clear and self-documenting
def calculate_mean(data):
    """Calculate the mean of a list of numbers."""
    return sum(data) / len(data)

def calculate_std(data):
    """Calculate the standard deviation of a list of numbers."""
    mean = calculate_mean(data)
    variance = sum((x - mean) ** 2 for x in data) / len(data)
    return variance ** 0.5

data = [45, 52, 48, 51, 49, 55, 47]
mean = calculate_mean(data)
std = calculate_std(data)
```

3. Make Code Easier to Share and Reuse

Functions can be collected into modules and shared:

```
# In file: neuroscience_utils.py
def calculate_firing_rate(spike_count, duration):
    """Calculate firing rate in Hz."""
    return spike_count / duration

def calculate_cv(isi_values):
    """Calculate coefficient of variation for inter-spike
intervals."""
    mean_isi = sum(isi_values) / len(isi_values)
    std_isi = (sum((x - mean_isi) ** 2 for x in isi_values) /
len(isi_values)) ** 0.5
    return std_isi / mean_isi

# In your analysis script:
from neuroscience_utils import calculate_firing_rate, calculate_cv

firing_rate = calculate_firing_rate(spike_count=150, duration=10.0)
```

3.3 Function Naming and Size

Good Function Names:

- Describe what the function does: `calculate_distance`, `get_user_input`, `is_valid`, `convert_to_celsius`
- Use verb-noun format: `load_data`, `save_results`, `plot_histogram`
- Follow PEP 8: lowercase with underscores

How Much Code in a Function? A function should do **one thing** and do it well. If you can't easily name the function or describe what it does in one sentence, it probably does too much.

```
        # TOO BROAD - does multiple unrelated things
        def analyze_everything(data):
            """This function loads data, processes it, analyzes it, and saves
results."""
            # 200 lines of code doing many different things...
            pass

        # BETTER - separate focused functions
        def load_spike_data(filename):
            """Load spike time data from CSV file."""
            pass

        def calculate_firing_rates(spike_times):
            """Calculate firing rates from spike times."""
            pass

        def plot_rate_histogram(firing_rates):
            """Plot histogram of firing rates."""
            pass

        def save_analysis_results(results, filename):
            """Save analysis results to JSON file."""
            pass
```

Important: In Python, functions must be defined **before** they are called:

```
        # WRONG - NameError
        greet("Ana")  # Function not yet defined!

        def greet(name):
            print(f"Hello, {name}")

        # CORRECT
        def greet(name):
            print(f"Hello, {name}")

        greet("Ana")  # Now it works
```

3.4 Function Parameters and Arguments

Terminology:

- **Parameters**: Variables listed in the function definition
- **Arguments**: Actual values passed when calling the function

```
        def calculate_distance(x1, y1, x2, y2):  # x1, y1, x2, y2 are
parameters
            """Calculate Euclidean distance between two points."""
            return ((x2 - x1) ** 2 + (y2 - y1) ** 2) ** 0.5

        # 10, 20, 30, 40 are arguments
        distance = calculate_distance(10, 20, 30, 40)
```

Positional Arguments

Arguments are matched to parameters by position:

```
        def describe_neuron(name, region, neuron_type, count):
            """Describe a neuron population."""
            print(f"Neuron: {name}")
            print(f"Brain region: {region}")
            print(f"Type: {neuron_type}")
            print(f"Count: {count}")

        # Arguments matched by position
        describe_neuron("Cell-001", "hippocampus", "pyramidal", 150)
        # name = "Cell-001", region = "hippocampus", etc.

        # WRONG ORDER = WRONG RESULTS
        describe_neuron(150, "pyramidal", "hippocampus", "Cell-001")
        # name = 150 (wrong!), region = "pyramidal" (wrong!), etc.
```

Keyword Arguments

You can specify which parameter each argument goes to:

```
        # Using keyword arguments - order doesn't matter
        describe_neuron(
            name="Cell-001",
            neuron_type="pyramidal",
            count=150,
            region="hippocampus"
        )

        # Mix positional and keyword (positional must come first!)
        describe_neuron("Cell-001", "hippocampus", neuron_type="pyramidal",
count=150)
```

Default Parameter Values

Parameters can have default values for common cases:

```
        def analyze_spike_train(spike_times, bin_size=1.0, window=None):
            """
            Analyze spike train data.

            Parameters:
            -----------
            spike_times : list
                List of spike times in seconds
            bin_size : float, optional
                Size of bins for histogram (default: 1.0 second)
            window : tuple, optional
                (start, end) time window to analyze (default: use all data)
            """
            if window is None:
                window = (min(spike_times), max(spike_times))

            # Analysis code here...
            print(f"Analyzing {len(spike_times)} spikes")
            print(f"Bin size: {bin_size} s")
            print(f"Window: {window[0]:.2f} to {window[1]:.2f} s")

        # Use default values
        analyze_spike_train([0.1, 0.5, 1.2, 2.3, 3.1])

        # Override bin_size only
        analyze_spike_train([0.1, 0.5, 1.2, 2.3, 3.1], bin_size=0.5)

        # Override both defaults
        analyze_spike_train([0.1, 0.5, 1.2, 2.3, 3.1], bin_size=0.5, window=
(0, 2.0))
```

Rules for default parameters:

- Must come after required parameters
- Evaluated once when function is defined (watch out with mutable defaults!)
- Use `None` as default for mutable types (lists, dicts):

```
        # DANGER - mutable default parameter!
        def add_spike(spike_time, spike_list=[]):  # BAD!
            spike_list.append(spike_time)
            return spike_list

        # Each call shares the same list!
        spikes1 = add_spike(1.0)  # [1.0]
        spikes2 = add_spike(2.0)  # [1.0, 2.0] - unexpected!

        # CORRECT - use None
        def add_spike(spike_time, spike_list=None):  # GOOD
```

```
    if spike_list is None:
        spike_list = []
    spike_list.append(spike_time)
    return spike_list

spikes1 = add_spike(1.0)  # [1.0]
spikes2 = add_spike(2.0)  # [2.0] - correct!
```

Variable Number of Arguments

Sometimes you don't know how many arguments you'll receive:

```
# *args collects extra positional arguments into a tuple
def calculate_mean(*values):
    """Calculate mean of any number of values."""
    if len(values) == 0:
        return 0
    return sum(values) / len(values)

mean1 = calculate_mean(5, 10, 15)           # 10.0
mean2 = calculate_mean(1, 2, 3, 4, 5)       # 3.0
mean3 = calculate_mean(42)                   # 42.0

# **kwargs collects extra keyword arguments into a dictionary
def create_neuron_record(neuron_id, region, **properties):
    """Create a neuron record with flexible properties."""
    record = {
        "id": neuron_id,
        "region": region
    }
    # Add all extra properties
    record.update(properties)
    return record

neuron1 = create_neuron_record("N001", "cortex",
                               cell_type="pyramidal",
                               layer="V",
                               firing_rate=45.2)

print(neuron1)
# {'id': 'N001', 'region': 'cortex', 'cell_type': 'pyramidal',
#  'layer': 'V', 'firing_rate': 45.2}
```

3.5 Returning Values

Functions can return values using the `return` statement:

```
def calculate_membrane_potential(current, resistance):
    """Calculate membrane potential using Ohm's law."""
    voltage = current * resistance
    return voltage
```

```
# Use returned value
v = calculate_membrane_potential(current=0.5, resistance=100)
print(f"Membrane potential: {v} mV")
```

Multiple Return Values

Python allows returning multiple values (actually returns a tuple):

```
def calculate_statistics(data):
    """Calculate mean and standard deviation."""
    n = len(data)
    mean = sum(data) / n
    variance = sum((x - mean) ** 2 for x in data) / n
    std = variance ** 0.5
    return mean, std  # Returns a tuple

# Unpack returned values
avg, sd = calculate_statistics([45, 52, 48, 51, 49])
print(f"Mean: {avg:.2f}, SD: {sd:.2f}")

# Or capture as tuple
stats = calculate_statistics([45, 52, 48, 51, 49])
print(f"Results: {stats}")  # (49.0, 2.449...)
```

Returning Different Types

Functions can return any Python object:

```
# Return boolean
def is_spiking(voltage, threshold=-55.0):
    """Check if voltage exceeds threshold."""
    return voltage > threshold

# Return list
def filter_spikes(spike_times, start, end):
    """Return spikes within time window."""
    return [t for t in spike_times if start <= t <= end]

# Return dictionary
def create_trial_data(trial_id, reaction_time, correct):
    """Create trial data dictionary."""
    return {
        "trial": trial_id,
        "rt": reaction_time,
        "correct": correct,
        "timestamp": "YYYY-MM-DD"
    }

# Return None (implicitly - no return statement)
def print_summary(data):
```

```
    """Print data summary (returns None)."""
    print(f"Number of trials: {len(data)}")
    print(f"Mean RT: {sum(data) / len(data):.2f}")
    # No return statement - implicitly returns None
```

Early Returns

The `return` statement immediately exits the function:

```
def validate_age(age):
    """Validate age input."""
    if age < 0:
        print("Error: Age cannot be negative")
        return False  # Exit here if invalid

    if age > 120:
        print("Error: Age seems unrealistic")
        return False  # Exit here if invalid

    print("Age is valid")
    return True  # Only reached if all checks pass

# The rest of function after return is never executed
def example():
    return 42
    print("This never prints!")  # Dead code!
```

3.6 Passing Functions as Arguments

Functions are objects in Python - you can pass them to other functions:

```
def rectangle_area(height, width):
    """Calculate rectangle area."""
    return height * width

def triangle_area(height, width):
    """Calculate triangle area."""
    return 0.5 * height * width

def circle_area(radius):
    """Calculate circle area."""
    return 3.14159 * radius ** 2

def calculate_total_area(shape_function, dimensions):
    """
    Calculate total area for multiple shapes.

    Parameters:
    -----------
    shape_function : function
        Function to calculate area (must accept 2 parameters)
```

```
    dimensions : list of tuples
        List of (height, width) tuples
    """
    total = 0
    for dim in dimensions:
        total += shape_function(dim[0], dim[1])
    return total

# Pass different functions to same code
rectangles = [(4, 5), (2, 1), (1, 10)]
triangles = [(4, 5), (2, 1), (1, 10)]

rect_area = calculate_total_area(rectangle_area, rectangles)    # 32
tri_area = calculate_total_area(triangle_area, triangles)        # 16

print(f"Total rectangle area: {rect_area}")
print(f"Total triangle area: {tri_area}")
```

Returning Functions:

```
def get_area_function(shape):
    """Return the appropriate area calculation function."""
    if shape == "rectangle":
        return rectangle_area
    elif shape == "triangle":
        return triangle_area
    else:
        return None

# Get and use the appropriate function
shape_type = "rectangle"
area_func = get_area_function(shape_type)

if area_func is not None:
    area = area_func(4, 5)
    print(f"{shape_type} area: {area}")
```

3.7 Scope and Namespace

Scope determines where variables are accessible in your code. Python has several levels of scope:

1. **Local scope** - inside a function
2. **Global scope** - at the module level
3. **Built-in scope** - Python's built-in names

```
# Global scope
global_var = "I'm global"
count = 0

def my_function():
    # Local scope
    local_var = "I'm local"
```

```
    count = 10  # This is a NEW local variable, doesn't affect global
count!

    print(global_var)   # Can read global variables
    print(local_var)    # Can read local variables
    print(count)        # Prints local count (10)

my_function()
print(global_var)       # Works - global
# print(local_var)      # ERROR - local_var doesn't exist here!
print(count)            # Prints global count (0) - unchanged!
```

The LEGB Rule - Python searches for names in this order:

1. **L**ocal - inside current function
2. **E**nclosing - inside enclosing functions
3. **G**lobal - at module level
4. **B**uilt-in - Python built-in names

```
x = "global x"

def outer():
    x = "outer x"

    def inner():
        x = "inner x"
        print(x)  # Prints "inner x" (Local)

    inner()
    print(x)      # Prints "outer x" (Enclosing)

outer()
print(x)          # Prints "global x" (Global)

# Built-in example
print(len([1, 2, 3]))  # len is a built-in function
```

Modifying Global Variables:

```
count = 0

def increment():
    global count  # Declare we want to modify global count
    count += 1

print(count)      # 0
increment()
print(count)      # 1
increment()
print(count)      # 2

# WITHOUT global keyword:
```

```
def bad_increment():
    count += 1    # ERROR! count not defined locally

# This raises: UnboundLocalError
```

Best Practice: Avoid modifying global variables - use function parameters and return values instead:

```
# BAD - modifies global
count = 0
def increment():
    global count
    count += 1

# GOOD - uses parameters and return
def increment(count):
    return count + 1

count = 0
count = increment(count)  # Clear that count is being updated
```

Warning about Namespace Pollution:

```
# Be careful with variable names in different scopes
data = [1, 2, 3, 4, 5]

def process_data():
    # This creates a NEW local variable, doesn't use global!
    data = []
    for i in range(10):
        data.append(i * 2)
    return data

result = process_data()
print(data)      # Still [1, 2, 3, 4, 5] - global unchanged
print(result)    # [0, 2, 4, 6, 8, 10, 12, 14, 16, 18] - local version
```

3.8 Recursive Functions

A **recursive function** calls itself to solve smaller versions of the same problem:

```
def factorial(n):
    """
    Calculate factorial of n using recursion.
    n! = n × (n-1) × (n-2) × ... × 2 × 1
    """
    # Base case - stop recursion
    if n <= 1:
        return 1

    # Recursive case - call itself with smaller problem
    return n * factorial(n - 1)
```

```
print(factorial(5))  # 5! = 5 × 4 × 3 × 2 × 1 = 120

# How it works:
# factorial(5) = 5 * factorial(4)
#              = 5 * (4 * factorial(3))
#              = 5 * (4 * (3 * factorial(2)))
#              = 5 * (4 * (3 * (2 * factorial(1))))
#              = 5 * (4 * (3 * (2 * 1)))
#              = 120
```

Critical: Always include a **base case** to stop recursion:

```
# DANGER - no base case!
def bad_countdown(n):
    print(n)
    return bad_countdown(n - 1)  # Never stops!

# CORRECT - has base case
def good_countdown(n):
    if n <= 0:
        print("Done!")
        return
    print(n)
    good_countdown(n - 1)

good_countdown(5)
# Output: 5, 4, 3, 2, 1, Done!
```

Practical Example - Fibonacci Sequence:

```
def fibonacci(n):
    """
    Calculate the nth Fibonacci number.
    Sequence: 0, 1, 1, 2, 3, 5, 8, 13, 21...
    """
    if n <= 0:
        return 0
    elif n == 1:
        return 1
    else:
        return fibonacci(n - 1) + fibonacci(n - 2)

for i in range(10):
    print(f"F({i}) = {fibonacci(i)}")
```

When to Use Recursion:

- ✅ Naturally recursive problems (trees, nested structures)
- ✅ Divide-and-conquer algorithms
- ✅ When code is clearer with recursion
- ❌ Simple loops are usually better for simple iteration

- ❌ Deep recursion can cause stack overflow errors

Summary and Key Takeaways

> [!summary] **Variables and Data Structures**
>
> - Variables store data and don't need explicit type declaration
> - Basic types: `int`, `float`, `str`, `bool`, `None`
> - Collections: `list` (mutable), `tuple` (immutable), `dict` (key-value), `set` (unique values)
> - Use `input()` for user inpVariables and Data Structures
> - Variables store data and don't need explicit type declaration
> - Basic types: `int`, `float`, `str`, `bool`, `None`
> - Collections: `list` (mutable), `tuple` (immutable), `dict` (key-value), `set` (unique values)
> - Use `input()` for user input, `print()` for output, f-strings for formatting ut, `print()` for output, f-strings for formatting

> [!summary] **Operators and Expressions**
>
> - **Arithmetic:** `+`, `-`, `*`, `/`, `//`, `%`, `**`
> - **Comparison:** `<`, `<=`, `>`, `>=`, `==` , `!=`
> - **Logical:** `and`, `or`, `not`
> - **Precedence:** arithmetic → comparison → logical (use parentheses for clarity!)

> [!summary] **Control Structures**
>
> - **`if-elif-else`:** Make decisions based on conditions
> - **`for` loops:** Iterate a known number of times or through sequences
> - **`while` loops:** Repeat while condition is true
> - **`break`:** Exit loop early
> - **`continue`:** Skip to next iteration
> - **Always watch for:** proper indentation, infinite loops, off-by-one errors

> [!summary] **Exception Handling**
>
> - Use `try-except` blocks to handle errors gracefully
> - Handle specific exception types separately
> - Common exceptions: `ValueError`, `ZeroDivisionError`, `TypeError`, `NameError`

> [!summary] **Functions**
>
> - **Define:** `def function_name(parameters):`
> - **Benefits:** Avoid repetition, improve readability, enable reuse
> - **Parameters:** Positional, keyword, default values, `*args`, `**kwargs`
> - **Return:** Use `return` to send values back
> - **Scope:** Local vs. global - be mindful of variable accessibility
> - **Recursion:** Functions can call themselves (need base case!)

> [!important] **Best Practices**
>
> 1. Use descriptive variable and function names
> 2. Follow PEP 8 style guidelines (4-space indentation, lowercase_with_underscores)
> 3. Write docstrings for functions
> 4. Keep functions focused on one task

> 5. Use comments to explain *why*, not *what*
> 6. Handle errors with try-except blocks
> 7. Test your code with different inputs

Simple Practice Exercises

Exercise 1: Temperature Converter

Write a program that:

1. Asks the user for a temperature
2. Asks whether it's in Celsius or Fahrenheit
3. Converts to the other unit and displays the result
4. Handles invalid input gracefully

Formulas:

- °F = (°C × 9/5) + 32
- °C = (°F - 32) × 5/9

Exercise 2: Spike Rate Analysis

Write a function `analyze_spike_train(spike_times, duration)` that:

1. Takes a list of spike times (in seconds) and recording duration
2. Calculates firing rate (spikes/second)
3. Calculates inter-spike intervals (ISI)
4. Returns mean ISI and coefficient of variation (CV = std/mean)

Exercise 3: Data Filtering

Write a function that takes a list of experimental measurements and:

1. Removes outliers (values > 3 standard deviations from mean)
2. Returns cleaned data and number of outliers removed
3. Handles empty lists gracefully

Exercise 4: Recursive List Sum

Write a recursive function to sum all numbers in a nested list:

```
# Example: [1, [2, 3], [[4], 5]] should return 15
```

Additional Resources

Official Python Documentation:

- Python Tutorial: https://docs.python.org/3/tutorial/
- Built-in Functions: https://docs.python.org/3/library/functions.html
- PEP 8 Style Guide: https://pep8.org/

Learning Resources:

- Real Python: https://realpython.com/
- Python Standard Library Reference: https://docs.python.org/3/library/
- Python Module of the Week: https://pymotw.com/3/

Practice:

- Python Challenge: http://www.pythonchallenge.com/
- HackerRank: https://www.hackerrank.com/domains/python
- Project Euler: https://projecteuler.net/

This handout accompanies the "Introduction to Scientific Programming" course at the Center for Neuroscience and Cell Biology, University of Coimbra. It provides comprehensive coverage of programming fundamentals but is not intended as a standalone introduction to Python. For complete beginners, we recommend supplementing with additional resources and hands-on practice.

Updated for Python 3.11+ with modern best practices